

PROJECT DESIGN PHASE-II

Solution Requirements Specification (SRS)

Date	18 June 2026	Project Name	ShopEz (MERN Full Stack)
Team ID	--	Maximum Marks	4 Marks
Student	INAPAKUTHIKA BHARATH VARMA (bharathvarma906@gmail.com)		

Functional Requirements (FR)

ID	Requirement (Epic)	Description & Logic
FR-1	User Onboarding - Input Forms - Validation	Functionality for new customers to establish accounts through email registration and encrypted passcode storage.
FR-2	Login Services - JWT Security - Session Lifecycle	A fortified sign-in architecture providing stateless user verification via JSON Web Tokens.
FR-3	Digital Catalog - Dynamic Data - Tiled Display	Real-time presentation of Men, Women, and Kids inventory categories sourced directly from MongoDB.
FR-4	Smart Search Keyword Analysis	Enables shoppers to query the product store using specific terms with instantaneous results.
FR-5	Faceted Filtering Price, Style, & Size	Live refinement of product lists using React state logic to avoid full-page reloads.
FR-6	Shopping Basket - Persisted State - CRUD	Facility to add, modify, or delete items within a persistent user cart.
FR-7	Order Finalization - Delivery Info - Totals	Guided checkout flow for entering shipping coordinates and reviewing cost breakdowns before purchase.
FR-8	Shipment Tracking Status Milestones	User portal to monitor the lifecycle (Processing/Shipped/Arrived) of current orders.
FR-9	Inventory Control - Media Mgmt - Pricing	Back-office dashboard for administrators to add, update, or archive store SKU items.
FR-10	Sales Oversight - Order Hub - Reporting	Unified admin center for global order fulfillment and monitoring commercial performance metrics.

Non-Functional Requirements (NFR)

ID	Attribute	Requirement Specification	Importance
NFR-1	Security	All user credentials must be salted and hashed via Bcrypt; API routes require JWT verification.	CRITICAL
NFR-2	Efficiency	Primary page rendering in under 2 seconds; search API latency should not exceed 500ms.	HIGH
NFR-3	Elasticity	The system must support horizontal expansion via MongoDB Atlas cluster partitioning.	HIGH
NFR-4	Responsive UI	Adaptive, mobile-first design ensuring consistent usability across phones, tablets, and desktops.	HIGH
NFR-5	Reliability	Targeting 99.9% platform uptime, particularly during high-traffic sales periods.	CRITICAL
NFR-6	Data Cohesion	Maintaining strict referential links between User accounts, Product data, and Transaction records.	CRITICAL